

STUDY AND LEARNING CENTRE

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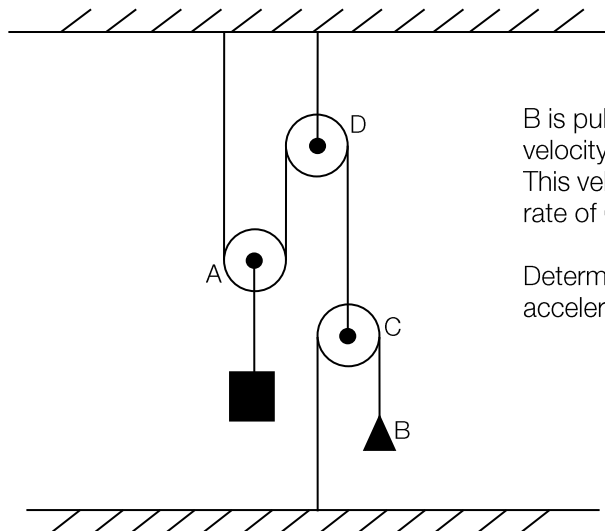
STUDY TIPS



WORKED SOLUTIONS

ENDY2.1 BLOCKS & PULLEYS

Question

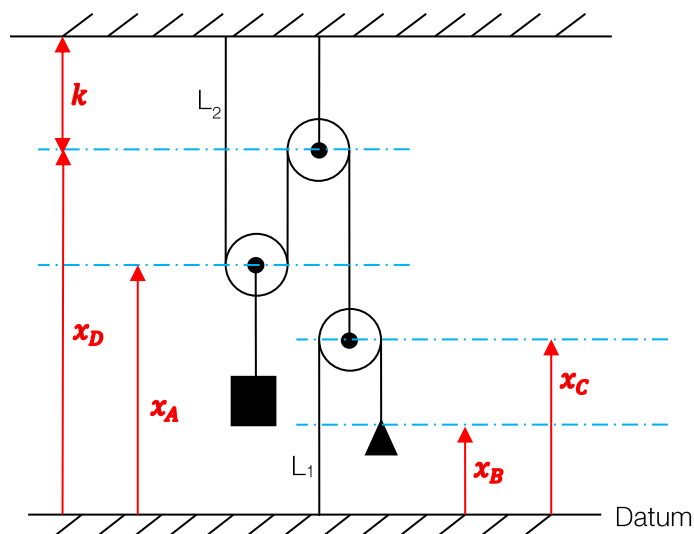


B is pulled downwards with a velocity of 1.2 ms^{-1} .
This velocity is decreasing at a rate of 0.6 ms^{-2} .

Determine the velocity and acceleration of A.

Worked Solution

Measure the position of all blocks from a common datum, and dimension on drawing. These dimensions are position vectors and direction is important.
Let $\uparrow = +\text{ve}$.



$$\dot{x}_B = -1.2 \text{ ms}^{-1}$$

$$\ddot{x}_B = 0.6 \text{ ms}^{-2}$$

Determine the length of the two ropes in terms of the position vectors.

$$\text{Eq}^{\text{n}} 1. \quad L_1 = 2x_C - x_B$$

$$\text{Eq}^{\text{n}} 2. \quad L_2 = 2(x_D - x_A) + (x_D - x_C) + k = -2x_A - x_C + 3x_D + k$$

From our diagram above we note that L_1 , L_2 , x_D and k are all constants.

Differentiate both sides of the equations above with respect to time.

$$\begin{aligned} \text{Eq}^{\text{n}} 1. \quad \frac{d}{dt}(L_1) &= \frac{d}{dt}(2x_C - x_B) \\ 0 &= 2\dot{x}_C - \dot{x}_B \\ \dot{x}_C &= \frac{\dot{x}_B}{2} = \frac{-1.2}{2} = -0.6 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \text{Eq}^{\text{n}} 2. \quad \frac{d}{dt}(L_2) &= \frac{d}{dt}(-2x_A - x_C + 3x_D + k) \\ 0 &= -2\dot{x}_A - \dot{x}_C + 3(0) \\ \dot{x}_A &= \frac{-\dot{x}_C}{2} = \frac{-(-0.6)}{2} = 0.3 \text{ ms}^{-1} \end{aligned}$$

Differentiating again yields:

$$\begin{aligned} 0 &= 2\ddot{x}_C - \ddot{x}_B \\ \ddot{x}_C &= \frac{\ddot{x}_B}{2} = \frac{0.6}{2} = 0.3 \text{ ms}^{-2} \\ 0 &= -2\ddot{x}_A - \ddot{x}_C + 3(0) \\ \ddot{x}_A &= \frac{-\ddot{x}_C}{2} = \frac{-(0.3)}{2} = -0.15 \text{ ms}^{-2} \end{aligned}$$

$$\therefore \text{velocity of A} = \dot{x}_A = 0.3 \text{ ms}^{-1}$$

$$\therefore \text{acceleration of A} = \ddot{x}_A = -0.15 \text{ ms}^{-2}$$